

Lem riemann lebesgue 11

Lema 1 (de Riemann-Lebesgue). Sea $f \in \mathcal{L}^1([0, 1))$, entonces

$$\lim_{|n| \rightarrow \infty} |\widehat{f}(n)| = 0.$$

Demostración: Partiendo de la identidad de Euler, tenemos que

$$\begin{aligned} \widehat{f}(n) &= \int_0^1 f(x) e^{-2\pi i n x} dx = - \int_0^1 f(x) e^{-2\pi i n x} e^{-\pi i} dx = - \int_0^1 f(x) e^{-2\pi i n (x+1/2n)} dx \\ &= - \int_{\frac{1}{2n}}^{1+\frac{1}{2n}} f\left(y - \frac{1}{2n}\right) e^{-2\pi i n y} dy = - \int_0^1 f\left(x - \frac{1}{2n}\right) e^{-2\pi i n x} dx. \\ \implies \widehat{f}(n) &= \frac{1}{2} \left[\int_0^1 f(x) e^{-2\pi i n x} dx - \int_0^1 f\left(x - \frac{1}{2n}\right) e^{-2\pi i n x} dx \right] \\ &= \frac{1}{2} \int_0^1 \left[f(x) - T_{1/2n} f(x) \right] e^{-2\pi i n x} dx \end{aligned}$$

Por el lem-convergencia-lp-traslacion/??, tenemos que

$$0 \leq \lim_{|n| \rightarrow \infty} |\widehat{f}(n)| \leq \lim_{|n| \rightarrow \infty} \frac{1}{2} \int_0^1 |f(x) - T_{1/2n} f(x)| dx = 0. \quad \blacksquare$$

Referenciado en

- prop-criterio-dini
- lem-serie-fourier-derivada

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